

Continuum Mechanics HW5

Daniel Haim Breger, 316136944

Technion

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1 Flamant Solution for a Point Tangential Stress

1.1 radial only

The stress Q is applied purely in the radial direction so none of it is applied in the angular direction, so $\sigma_{\theta r} = \sigma_{\theta\theta} = 0$.

1.2 Airy stress

Since the solutions we need are equilibrium solutions, we have $\nabla \cdot \sigma = 0$. Therefore integrating over half a sphere like in the TA session

$$0 = \int_A dA \nabla \cdot \sigma = \int_0^R dr \int_{-\pi/2}^{\pi/2} d\theta r \nabla \cdot \sigma \quad (1)$$

$$= \int_{-\pi/2}^{\pi/2} d\theta R \hat{\mathbf{r}} \cdot \sigma + \int_{-R}^R (-\hat{\mathbf{x}}) \cdot \sigma dy \quad (2)$$

$$= \int_{-\pi/2}^{\pi/2} d\theta R \sigma_{rr} \hat{\mathbf{r}} - \int_{-R}^R \sigma_{xy} \hat{\mathbf{y}} dy \quad (3)$$

$$= \int_{-\pi/2}^{\pi/2} d\theta R \sigma_{rr} \hat{\mathbf{r}} + Q \int_{-R}^R \delta(r) \hat{\mathbf{y}} dy \quad (4)$$

so

$$Q \int_{-R}^R \delta(r) \hat{\mathbf{y}} dy = - \int_{-\pi/2}^{\pi/2} d\theta R \sigma_{rr} \hat{\mathbf{r}} \quad (5)$$

$$= -R \left(\int_{-\pi/2}^{\pi/2} d\theta \sigma_{rr} \cos \theta \hat{\mathbf{x}} + \int_{-\pi/2}^{\pi/2} d\theta \sigma_{rr} \sin \theta \hat{\mathbf{y}} \right) \quad (6)$$

which we can now split into two equations by unit vector

$$Q = -R \int_{-\pi/2}^{\pi/2} d\theta \sigma_{rr} \sin \theta \quad (7)$$

$$0 = R \int_{-\pi/2}^{\pi/2} \sigma_{rr} \cos \theta \quad (8)$$

Since Q is a constant the integral must resolve to a quantity that includes $1/r$ in it somewhere, meaning

$$\sigma_{rr} \propto \frac{F(\theta)}{r} \quad (9)$$

where F is some function. Using the expression from the training

$$\sigma_{rr} = \frac{1}{r} \frac{\partial \chi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \chi}{\partial \theta^2} \propto \frac{1}{r} \implies F(\theta) = \frac{\partial \chi}{\partial r} + \frac{1}{r} \frac{\partial^2 \chi}{\partial \theta^2} \quad (10)$$

and continuing with the training we find that we must have

$$\chi = (A \cos \theta + B \sin \theta) r \theta \quad (11)$$

and now derivatives

$$\sigma_{rr} = \frac{1}{r} \frac{\partial \chi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \chi}{\partial \theta^2} \quad (12)$$

$$= \dots = \frac{-2A \sin \theta + 2B \cos \theta}{r} \quad (13)$$

again from the training. Throwing this into the expressions we found

$$0 = R \int_{-\pi/2}^{\pi/2} \frac{-2A \sin \theta + 2B \cos \theta}{R} \cos \theta \quad (14)$$

$$= \int_{-\pi/2}^{\pi/2} (-2A \sin \theta + 2B \cos \theta) \cos \theta \quad (15)$$

$$= \int_{-\pi/2}^{\pi/2} 2B \cos^2 \theta \implies B = 0 \quad (16)$$

and

$$Q = 2A \int_{-\pi/2}^{\pi/2} d\theta \sin^2 \theta \quad (17)$$

$$= 2A \cdot \pi/2 = \pi A \implies A = \frac{Q}{\pi} \quad (18)$$

so in total

$$\chi = \frac{Q}{\pi} \cos \theta r \theta, \quad \sigma_{rr} = -\frac{2Q}{\pi} \frac{\sin \theta}{r} \quad (19)$$

1.3 equal-stress contours

for constant stress

$$\frac{2Q}{\pi} \frac{\sin \theta}{r} \quad (20)$$

so as in the training we define

$$D = \frac{Q}{\pi C} \quad (21)$$

and

$$\frac{1}{D} = \frac{2 \sin \theta}{r^2} r^2 = 2Dr \sin \theta \quad (22)$$

converting to cartesian for humans

$$x^2 + y^2 - 2Dy = 0 \implies x^2 + y^2 - 2Dy + D^2 = D^2 \implies x^2 + (y - D)^2 = D^2 \quad (23)$$

giving a circle around $y = D$. For the other circle, we notice that setting $D \rightarrow -D$ changes nothing, so $y = -D$ is also a circle center.

1.4 general tangential stress

in cartesian from radial

$$\sigma_{xx} = \sigma_{rr} \cos^2 \theta = \frac{2Q}{\pi} \frac{x^2 y}{(x^2 + y^2)^2} \quad (24)$$

$$\sigma_{yy} = \sigma_{rr} \sin^2 \theta = \frac{2Q}{\pi} \frac{y^3}{(x^2 + y^2)^2} \quad (25)$$

$$\sigma_{xy} = \sigma_{rr} \sin \theta \cos \theta = \frac{2Q}{\pi} \frac{xy^2}{(x^2 + y^2)^2} \quad (26)$$

but since we want a general stress distribution, instead of Q we have $q(s)$ so

$$\sigma_{xx} = \frac{2x^2}{\pi} \int_{-\infty}^{\infty} \frac{(y-s) q(s)}{(x^2 + (y-s)^2)^2} \quad (27)$$

$$\sigma_{yy} = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{(y-s)^3 q(s)}{(x^2 + (y-s)^2)^2} \quad (28)$$

$$\sigma_{xy} = \frac{2x}{\pi} \int_{-\infty}^{\infty} \frac{(y-s)^2 q(s)}{(x^2 + (y-s)^2)^2} \quad (29)$$

2 Model with Cylindrical Hole Under Uniaxial Tension

2.1 superposition

We can write

$$\sigma_{yy} = a\sigma_1 + b\sigma_2 \stackrel{!}{=} \sigma_\infty, \quad \sigma_{xx} = a\sigma_1 - b\sigma_2 \stackrel{!}{=} 0 \quad (30)$$

because of the directions of the tension, and this requires

$$a = b = \frac{1}{2} \quad (31)$$

2.2 validity

Since we're working with $L \gg R$, the domain's exact shape is effectively irrelevant, since the boundary conditions we produce from it are treated to be at infinity. Looking at the boundary conditions at infinity as defined in cartesian or cylindrical coordinates produces the same boundary conditions.

2.3 symmetric tension solution

we know

$$\sigma_{xx} = \frac{\partial^2 \chi}{\partial x^2}, \quad \sigma_{yy} = \frac{\partial^2 \chi}{\partial y^2}, \quad \sigma_{xy} = -\frac{\partial^2 \chi}{\partial x \partial y} \quad (32)$$

and our boundary conditions are

$$\sigma_{xx}(r \gg R) = -\sigma_2 \quad (33)$$

$$\sigma_{yy}(r \gg R) = \sigma_2 \quad (34)$$

$$\sigma_{xy}(r \gg R) = 0 \quad (35)$$

thus

$$\frac{\partial^2 \chi(r \gg R)}{\partial x^2} = -\sigma_2 \implies \chi(r \gg R) = \frac{\sigma_2}{2} x^2 + A(y)x + B(y) \quad (36)$$

$$\frac{\partial^2 \chi}{\partial y^2} = \sigma_2 \implies \chi(r \gg R) = -\frac{\sigma_2}{2} y^2 + C(x)y + D(x) \quad (37)$$

combining both we get

$$\chi = \frac{\sigma_2}{2} (x^2 - y^2) + Axy \quad (38)$$

using the condition for xy we get $A = 0$ so

$$\chi = \frac{\sigma_2}{2} (x^2 - y^2) = \frac{\sigma_2}{2} r^2 (\cos^2 \theta - \sin^2 \theta) = \frac{\sigma_2}{2} r^2 \cos 2\theta \quad (39)$$

using the Michell solution

$$\chi = (Ar^2 + Br^{-2} + Cr^4 + D) \cos 2\theta \quad (40)$$

we can immediately say that $C = 0$ and $A = \frac{\sigma_2}{2}$. To finish we need to consider the boundary conditions at the edge of the hole

$$\sigma_{rr}(r = R) = \sigma_{r\theta}(r = R) = 0 \quad (41)$$

and we find them from the airy stress

$$\sigma_{rr} = \frac{1}{r} \frac{\partial \chi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \chi}{\partial \theta^2} = (\sigma_2 - 2Dr^{-4}) \cos 2\theta - 4 \left(\frac{\sigma_2}{2} + Cr^{-2} + Dr^{-4} \right) \cos 2\theta \quad (42)$$

$$\sigma_{r\theta} = -\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial \chi}{\partial \theta} \right) = \frac{\partial}{\partial r} (\sigma_2 r + 2Cr^{-1} + 2Dr^{-3}) \sin 2\theta = (\sigma_2 - 2Cr^{-2} + Dr^{-4}) \sin 2\theta \quad (43)$$

now using the boundary conditions

$$\sigma_2 - 2CR^{-2} - 6DR^{-4} = 0 \quad (44)$$

$$\sigma_2 + 6DR^{-4} + 4CR^{-2} = 0 \quad (45)$$

which gives

$$C = -\sigma_2 R^2, \quad D = \frac{\sigma_2}{2} R^4 \quad (46)$$

so

$$\chi = \frac{\sigma_2}{2} R^2 \left(w \frac{r^2}{R^2} + \frac{R^2}{r^2} - 2 \right) \cos 2\theta \quad (47)$$

2.4 general solution

We'll combine the solutions of the two cases we know. For subfigure b

$$\sigma_{rr} = \sigma_1 \left(1 - \frac{R^2}{r^2} \right), \quad \sigma_{\theta\theta} = \sigma_1 \left(1 + \frac{R^2}{r^2} \right), \quad \sigma_{r\theta} = 0 \quad (48)$$

and for subfigure c

$$\sigma_{rr} = -\sigma_2 \left(1 - 4\frac{R^2}{r^2} + 3\frac{R^4}{r^4} \right) \cos 2\theta, \quad \sigma_{\theta\theta} = \sigma_2 \left(1 + 3\frac{R^4}{r^4} \right) \cos 2\theta, \quad \sigma_{r\theta} = \sigma_2 \left(1 + 2\frac{R^2}{r^2} - 3\frac{R^4}{r^4} \right) \sin 2\theta \quad (49)$$

inserting $\sigma_1 = \sigma_2 = \frac{1}{2}\sigma_\infty$

$$\sigma_{rr} = \frac{\sigma_\infty}{2} \left((1 - \cos 2\theta) + (4 \cos 2\theta - 1) \frac{R^2}{r^2} - 3 \frac{R^4}{r^4} \cos 2\theta \right) \quad (50)$$

$$\sigma_{r\theta} = \frac{\sigma_\infty}{2} \left(1 + \frac{R^2}{r^2} - 3 \frac{R^4}{r^4} \right) \sin 2\theta \quad (51)$$

$$\sigma_{\theta\theta} = \frac{\sigma_\infty}{2} \left((1 + 2 \cos 2\theta) + \frac{R^2}{r^2} + 3 \frac{R^4}{r^4} \cos 2\theta \right) \quad (52)$$

2.5 stress at edge of hole

plugging in $r = R$

$$\sigma_{rr} = \frac{\sigma_\infty}{2} ((1 - \cos 2\theta) + (r \cos 2\theta - 1) - 3 \cos 2\theta) = 0 \quad (53)$$

$$\sigma_{r\theta} = \frac{\sigma_\infty}{2} (1 + 2 - 3) \sin 2\theta = 0 \quad (54)$$

$$\sigma_{\theta\theta} = \frac{\sigma_\infty}{2} (1 + 2 \cos 2\theta + 1 + 3 \cos 2\theta) = \frac{\sigma_\infty}{2} (2 + 5 \cos 2\theta) \quad (55)$$

for the maximum and minimum

$$\frac{d\sigma_{\theta\theta}}{d\theta} = -5\sigma_{\infty} \sin 2\theta = 0 \implies \theta = \frac{\pi k}{2} \quad (56)$$

so

$$\sigma_{\theta\theta \max} = \frac{7}{2}\sigma_{\infty}, \quad \sigma_{\theta\theta \min} = -\frac{3}{2}\sigma_{\infty} \quad (57)$$

the maximum stress experienced is simply

$$\sigma_{\max} = \frac{7}{2}\sigma_c \quad (58)$$

3 Finite Cylinder

3.1 stress tensor

Since $\chi = C\theta$ and by definition

$$\sigma_{rr} = 0, \quad \sigma_{\theta\theta} = 0, \quad \sigma_{r\theta} = -\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial \chi}{\partial \theta} \right) = \frac{C}{r^2} \quad (59)$$

3.1.1 boundary conditions

Unless I'm missing something, we just plug in $r = R_{1/2}$

$$\sigma_{rr} = 0, \quad \sigma_{\theta\theta} = 0 \quad (60)$$

and

$$\sigma_{r\theta}(R_1, \theta) = \frac{C}{R_1^2}, \quad \sigma_{r\theta}(R_2, \theta) = \frac{C}{R_2^2} \quad (61)$$

3.2 meaning

using the hint

$$\mathbf{T} = \int_{\partial V} dS \mathbf{r} \times \boldsymbol{\sigma} \quad (62)$$

plugging in the tensor we have

$$\mathbf{T} = \int_{\partial V} dS r \sigma_{r\theta} \hat{\mathbf{z}} \quad (63)$$

$$= \int_0^{2\pi} \int_0^L dz d\theta R^2 \frac{C}{R^2} \hat{\mathbf{z}} = 2\pi LC \hat{\mathbf{z}} \quad (64)$$

The torque is constant regardless of the radius at which it's calculated, but we can tell it'll have opposite signs because the direction of the normal would change, so the total torque applied to the cylinder will be 0.